Project 1

STAT 801A, Fall 2026

Due on January 1, 2026 amt 9:00 P1

Overview

The American Community Survey is a large national survey that collects information on many aspects of life in the United States. You are tasked with analyzing a subset of the 2022 survey for people who live and work in Lancaster County, Nebraska. In this particular subset, respondents were asked how long it takes for them to commute to work and only responses for people who drive alone are included. The data is available in the file `travel-time-31109.csv`.

Variable Descriptions:

- `RESPONDENT`: Unique identifier for each respondent
- `TRANTIME`: Travel time to work in minutes
- `DEPARTS`: Time of day the respondent leaves for work (e.g., 7:30 AM = 730)
- `ARRIVES`: Time of day the respondent arrives at work (e.g., 7:30 AM = 730)

Problems

Complete each of the following problems; solutions and brief explanations are provided below. The solution code reads `travel-time-31109.csv` from the project folder.

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

Solution: Problem 1

```r
# Read the data and show the first 10 rows
ne_work_commute <- read.csv("travel-time-31109.csv")
head(ne_work_commute, 10)
```

```
   RESPONDENT TRANTIME DEPARTS ARRIVES
1           1        7     647     654
2           2        8    1305    1309
3           3       10    1445    1454
4           4       10    1605    1614
5           5       10    1605    1614
6           6       15    1515    1534
7           7       15    1435    1449
8           8        7    1545    1554
9           9       12    1535    1544
10         10       15    1435    1449
```

Solution: Problem 2

```r
# Get dimensions and display a one-sentence summary
dim_ne <- dim(ne_work_commute)
dim_ne
```

```
[1] 861   4
```

```r
paste0("The dataset has ", dim_ne[1], " rows and ", dim_ne[2], " columns.")
```

```
[1] "The dataset has 861 rows and 4 columns."
```

Solution: Problem 3

```r
# Mean of TRANTIME
mean_tran <- mean(ne_work_commute$TRANTIME, na.rm = TRUE)
mean_tran
```

```
[1] 18.34727
```

Solution: Problem 4

```r
# Median of TRANTIME
median_tran <- median(ne_work_commute$TRANTIME, na.rm = TRUE)
median_tran
```

```
[1] 15
```

Solution: Problem 5

```r
# Variance of TRANTIME
var_tran <- var(ne_work_commute$TRANTIME, na.rm = TRUE)
var_tran
```

```
[1] 193.3293
```

Solution: Problem 6

```r
# Standard deviation of TRANTIME
sd_tran <- sd(ne_work_commute$TRANTIME, na.rm = TRUE)
sd_tran
```

```
[1] 13.90429
```

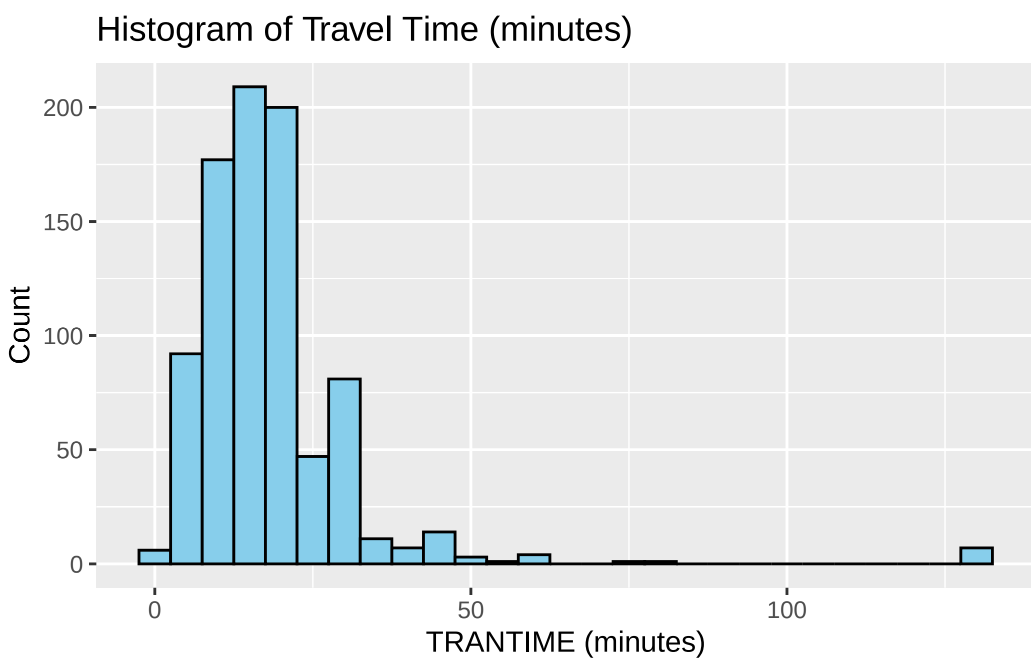
Solution: Problem 7

```r
# Quartiles / summary of TRANTIME
summary(ne_work_commute$TRANTIME)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   1.00   10.00   15.00   18.35   20.00  129.00 
```

Solution: Problem 8

```r
# Histogram of TRANTIME using ggplot2
ggplot(ne_work_commute, aes(x = TRANTIME)) +
  geom_histogram(binwidth = 5, color = "black", fill = "skyblue") +
  labs(title = "Histogram of Travel Time (minutes)", x = "TRANTIME (minutes)", y = "Count")
```



Solution: Problem 9

```r
# Compare mean and median and comment on which is preferable
mean_tran
```

```
[1] 18.34727
```

```r
median_tran
```

```
[1] 15
```

The mean is compared to the median above; if the mean is substantially larger than the median the distribution is right-skewed and the median is the more robust (preferred) measure of center.

Solution: Problem 10

```
# Compute IQR-based bounds and create filtered_travel_time (non-outliers)
q1 <- quantile(ne_work_commute$TRANTIME, 0.25, na.rm = TRUE)
q3 <- quantile(ne_work_commute$TRANTIME, 0.75, na.rm = TRUE)
iqr_val <- IQR(ne_work_commute$TRANTIME, na.rm = TRUE)
lower <- q1 - 1.5 * iqr_val
upper <- q3 + 1.5 * iqr_val
filtered_travel_time <- ne_work_commute$TRANTIME[ne_work_commute$TRANTIME >= lower & ne_work_
length(filtered_travel_time)
```

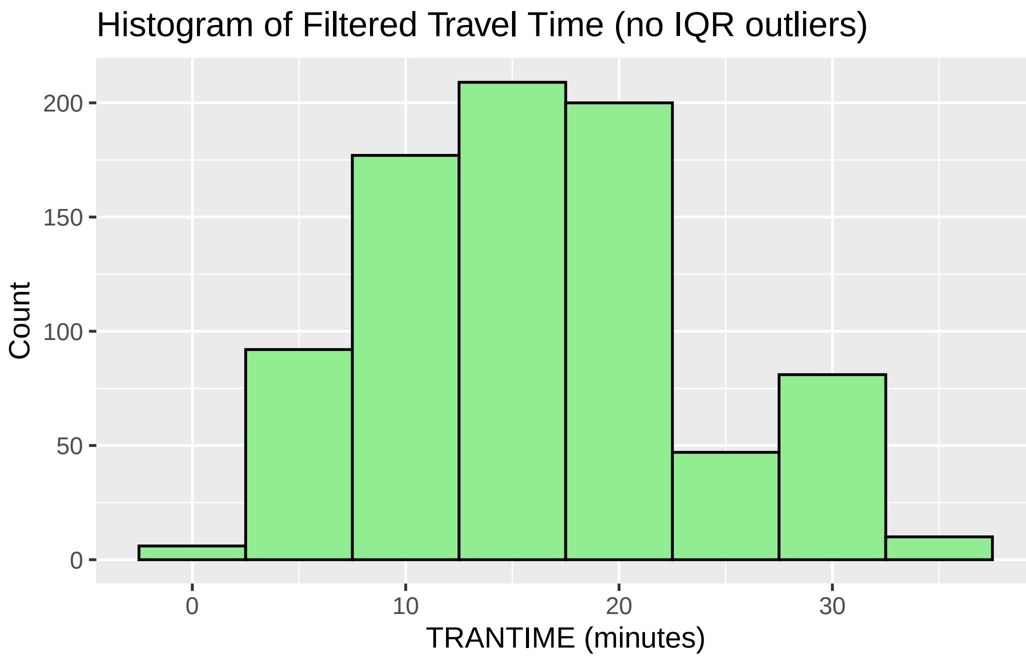
```
[1] 822
```

```
head(filtered_travel_time)
```

```
[1]  7  8 10 10 10 15
```

Solution: Problem 11

```
# Histogram of filtered_travel_time
filtered_df <- tibble(TRANTIME = filtered_travel_time)
ggplot(filtered_df, aes(x = TRANTIME)) +
  geom_histogram(binwidth = 5, color = "black", fill = "lightgreen") +
  labs(title = "Histogram of Filtered Travel Time (no IQR outliers)", x = "TRANTIME (minutes)
```



Solution: Problem 12

```
# Summary statistics for filtered_travel_time and formatted table
mean_f <- mean(filtered_travel_time, na.rm = TRUE)
median_f <- median(filtered_travel_time, na.rm = TRUE)
var_f <- var(filtered_travel_time, na.rm = TRUE)
sd_f <- sd(filtered_travel_time, na.rm = TRUE)

summary_table <- tibble(
  Statistic = c("Mean", "Median", "Variance", "SD"),
  Value = c(mean_f, median_f, var_f, sd_f)
) %>%
  mutate(Value = round(Value, 2))

knitr::kable(summary_table, caption = "Summary statistics for filtered_travel_time (rounded t
```

Table 1: Summary statistics for filtered_travel_time (rounded to 2 decimals)

Statistic	Value
Mean	16.24
Median	15.00

Statistic	Value
Variance	54.74
SD	7.40